

Pleural Disease — Fluid Analysis, Thoracentesis & Chest Tube



1. pH/glucose gauge

WHAT YOU SEE

Large dial showing pH in DANGER zone and GLUCOSE scale

WHAT IT ENCODES

Pleural fluid pH is the single most useful chemistry for deciding drainage of a parapneumonic effusion. Complicated parapneumonic effusion / empyema markers: pH < 7.20, glucose < 60 mg/dL, and LDH > 1000 IU/L. These reflect active infection/inflammation consuming glucose and producing lactic acid in a walled-off, poorly perfused pleural space. pH is more sensitive than glucose for predicting the need for tube drainage.

BOARD TRAP

A parapneumonic effusion with pH < 7.20 mandates chest tube drainage even if there is no frank pus on the tap. The trap is treating it with antibiotics alone because the fluid 'looks clear' — the chemistry, not the appearance, drives the drainage decision.

2. Rule of 7-6

WHAT YOU SEE

Banner 'RULE OF 7-6'

WHAT IT ENCODES

Memory hook: pH 7.2 and glucose 60 are the action thresholds for draining a parapneumonic effusion (pH < 7.2 OR glucose < 60 → drain). Pair this with the size/loculation rule: effusions that are large (> half the hemithorax), loculated, or with thickened parietal pleura also favor drainage regardless of chemistry.

BOARD TRAP

Don't memorize the number without the context — pH must be measured properly (ABG syringe, on ice, run promptly). A spuriously high pH from a delayed/air-exposed sample can push you below the threshold and lead you to wrongly withhold a needed chest tube.

3. Empyema = pus

WHAT YOU SEE

Jar of turbid/purulent fluid labeled COMPLICATED

WHAT IT ENCODES

Empyema = frank pus in the pleural space, a positive Gram stain/culture, or loculation. Pathophysiology progresses through 3 stages: exudative (free-flowing) → fibrinopurulent (loculating, fibrin strands) → organizing (thick peel/trapped lung). Most common organisms: Strep (S. pneumoniae, milleri/anginosus group), Staph aureus, and anaerobes (especially with aspiration/poor dentition). Empyema requires source control via drainage.

BOARD TRAP

Empyema is NEVER managed with antibiotics alone — it always needs drainage (chest tube ± intrapleural tPA/DNase, escalating to VATS). The trap is escalating/prolonging antibiotics for a non-resolving 'pneumonia' when the real problem is undrained pus; antibiotics penetrate the pleural space poorly.

4. Simple parapneumonic

WHAT YOU SEE

Jar of clear straw fluid

WHAT IT ENCODES

Uncomplicated (simple) parapneumonic effusion: free-flowing, small (< half hemithorax), pH > 7.2, glucose normal, LDH < 1000, and a negative Gram stain/culture. This represents sterile sympathetic exudate from adjacent pneumonia and resolves with antibiotics directed at the pneumonia alone.

BOARD TRAP

A small, free-flowing effusion with pH > 7.2 does NOT need a chest tube — the trap is over-treating with a drain. Manage with antibiotics + observation and re-image if the patient fails to improve, since a simple effusion can evolve into a complicated one.

5. 10 mL minimum tap

WHAT YOU SEE

Graduated cylinder marked 10 mL

WHAT IT ENCODES

A diagnostic thoracentesis should send fluid for: cell count with differential, total protein, LDH, glucose, pH, cytology, and Gram stain/culture (plus targeted tests like ADA, triglycerides, hematocrit, amylase as indicated). Lymphocyte-predominant fluid suggests TB or malignancy; neutrophil-predominant suggests acute parapneumonic/pancreatitis; eosinophilia (>10%) suggests air or blood in the pleural space.

BOARD TRAP

Any new effusion of uncertain cause warrants a diagnostic tap — the trap is reflexively treating presumed 'CHF effusion' with diuretics without sampling. Tap atypical effusions (unilateral, fever, pleuritic pain, or failure to resolve) to avoid missing infection or malignancy.

6. Send pH on ice + heparin

WHAT YOU SEE

Tubes labeled pH ICE + HEPARIN, cell count, cytology

WHAT IT ENCODES

Specimen handling determines accuracy: pH must be collected anaerobically in a heparinized ABG syringe, placed on ice, and analyzed on a blood-gas machine promptly. Cell count goes in EDTA; the largest volume goes for cytology (more fluid = higher yield, ideally 50-150 mL). Residual air in the syringe or delay allows CO2 to off-gas and falsely raises pH.

BOARD TRAP

Reporting pleural pH from a urine dipstick or a sample left at room temperature is unreliable — the trap is acting on a falsely normal pH and withholding drainage. Also, lidocaine contamination falsely LOWERS pH; flush the syringe and avoid drawing through residual anesthetic.

7. PPE / sterile tap

WHAT YOU SEE

Worker in simple PPE doing the tap on a patient's back

WHAT IT ENCODES

Ultrasound-guided thoracentesis is standard of care — it lowers pneumothorax and dry-tap rates and confirms a safe pocket. The posterior approach enters one to two intercostal spaces below the top of the effusion, and the needle should pass just ABOVE the upper border of the rib.

BOARD TRAP

Enter just ABOVE the lower rib (over the top of the rib below) to avoid the neurovascular bundle (vein/artery/nerve) that runs along the INFERIOR margin of each rib. The trap answer is going below the rib or skipping ultrasound — both increase the risk of bleeding and pneumothorax.

8. Don't drain too much

WHAT YOU SEE

Caution sign with lungs; 'AIR AFTER TAP = TRAPPED LUNG CLUE'

WHAT IT ENCODES

Limit a single therapeutic drainage to roughly 1.0-1.5 L (some monitor pleural manometry instead) and STOP for chest pain, persistent cough, or marked breathlessness — these herald re-expansion pulmonary edema, a non-cardiogenic edema of the rapidly re-expanded lung.

BOARD TRAP

Removing a large effusion all at once to 'make the patient comfortable' risks re-expansion pulmonary edema — the trap is draining 3 L in one sitting. Drain to symptoms or volume limit, not to dryness. Also, post-tap dyspnea with a hydropneumothorax may be trapped lung, not just procedural air.

9. Negative pressure = confirm

WHAT YOU SEE

Label 'NEGATIVE PRESSURE → CONFIRM'

WHAT IT ENCODES

Pleural manometry during drainage detects a non-expandable (trapped) lung: a rapid, steeply negative drop in pleural pressure (e.g., < -20 cm H₂O) as fluid is removed means the visceral pleura cannot re-expand to fill the space. This predicts that pleurodesis will fail and an indwelling catheter is preferable.

BOARD TRAP

Continuing to drain against rapidly falling negative pressure causes chest discomfort and risks re-expansion edema/pneumothorax ex vacuo — the trap is forcing out more fluid. Stop and recognize trapped lung instead of attributing the failed expansion to a procedural complication.

10. Small-bore tube OK

WHAT YOU SEE

Sign 'SMALL BORE 10-14 Fr usually sufficient'

WHAT IT ENCODES

Small-bore catheters (10-14 Fr) placed under image guidance are first-line and as effective as large-bore tubes for most parapneumonic effusions and empyemas, with less pain. Combine with intrapleural fibrinolytics (tPA + DNase) when fluid is loculated or thick.

BOARD TRAP

You do NOT need a large-bore tube for most empyemas — the trap is jumping straight to a 32 Fr surgical chest tube. Small-bore + tPA/DNase is preferred; reserve large bore for very viscous pus or hemothorax (clot).

11. Large bore for thick pus

WHAT YOU SEE

Sign 'LARGE BORE 24-36 Fr for thick pus'

WHAT IT ENCODES

Reserve large-bore tubes (24-36 Fr) for very viscous/thick pus that clogs small catheters and for hemothorax, where draining clotted blood and monitoring output (e.g., > 1500 mL initially or > 200 mL/hr) guides the decision for surgical exploration.

BOARD TRAP

Defaulting to large bore for every effusion adds pain and morbidity without benefit for thin fluid — the trap is over-sizing. Match tube caliber to fluid viscosity: thin → small bore; thick pus/blood → large bore.

12. tPA + DNase together

WHAT YOU SEE

Locked vault; 'tPA ALONE □ DNase ALONE □ BOTH TOGETHER □'

WHAT IT ENCODES

For loculated/complicated parapneumonic effusion or empyema, intrapleural tPA (breaks fibrin septations) PLUS DNase (lowers fluid viscosity by digesting DNA) given together improves drainage, shortens hospital stay, and reduces surgical referral (MIST2 trial). Typical regimen: tPA 10 mg + DNase 5 mg, instilled twice daily for 3 days.

BOARD TRAP

tPA alone or DNase alone does NOT work — only the COMBINATION is effective (MIST2 showed monotherapy arms failed). The trap is selecting 'intrapleural fibrinolytic' as a single agent; the board answer is the dual tPA + DNase regimen.

13. Loculated → continue tPA

WHAT YOU SEE

Label 'STILL LOCULATED → CONTINUE tPA'

WHAT IT ENCODES

If imaging shows persistent loculation despite a well-positioned tube, complete the full course of intrapleural tPA + DNase (e.g., reassess at 48-72 hours) before escalating to surgery. Many patients clear loculations and avoid VATS with adequate fibrinolytic therapy.

BOARD TRAP

Rushing to surgery after one dose of fibrinolytics, or conversely flailing with an undrained tube for a week — both are traps. Give the complete fibrinolytic course and reassess with imaging; escalate to VATS only after documented failure.

14. VATS / decortication

WHAT YOU SEE

Surgeon at table peeling pleura; 'VATS', 'PLEURAL PEEL'

WHAT IT ENCODES

When tube drainage + tPA/DNase fail, or in organized (stage 3) empyema with a thick fibrous peel trapping the lung, the answer is surgery: VATS for debridement/early decortication, or open decortication for an established peel. Surgery removes the rind so the lung can re-expand.

BOARD TRAP

Surgery (decortication) is the answer when tube + tPA/DNase fail or the lung is trapped by an organized peel — the trap is continuing to cycle fibrinolytics indefinitely into an organized empyema that fibrinolytics can no longer dissolve. Recognize the organizing stage and refer to surgery.